

How Fast Does Coordination Value Decay?

Exact Delayed-Pooling Laws for Gaussian Teams

Scott Moeller and Bhaskar Krishnamachari

Abstract—Fresh local measurements can yield better team decisions than a delayed pool containing every sensor’s data. We derive the exact delay at which this reversal occurs in a Gaussian model with a common temporal decay rate. Agents track a shared changing signal, with a cost for tracking error and disagreement. When agents keep using current local measurements, each needs only three numbers: its current reading, its reading at the pooling time, and the pooled average. These attain the same optimal performance as sharing all sensor histories through the pooling time while retaining current local observations. We quantify the value of keeping the old local reading and show that the pooling gain decays exponentially with age. The law extends to unequal sensor quality and independent signal components with different temporal rates. It also gives an explicit communication-price threshold, or equivalent latency threshold, beyond which no refreshing is optimal. Below the price threshold, the optimal periodic interval is unique. Periodic refreshing is optimal among all schedules independent of the sensor processes. Every result is proved within the letter.

Keywords: Decentralized control, sensor fusion, control system architecture.

I. INTRODUCTION

Sharing every sensor’s data can lead to worse team decisions than acting locally. Collecting and distributing the data takes time, and a changing environment can make a broad view outdated before it is used. We identify the exact delay beyond which fresh local measurements outperform a complete delayed observation pool. Both decision rules are optimized for their available information. In an eight-sensor example, local operation has about 26% lower team loss than decisions based on a pool one coherence time old, before any communication charge is included.

We consider agents tracking a common changing signal through noisy local sensors. Their shared objective penalizes tracking error and disagreement among their actions. The signal and sensor errors evolve as Gaussian processes with a common temporal decay rate. Agents can act on current local readings, use a delayed pool, or combine both. For the combined architecture, we prove that each agent needs just its current reading, its reading at the last pooling time, and the average of all sensors’ readings at that time. Those three numbers achieve the same minimum team loss as giving each agent every sensor’s history up to the pooling time, together with its current local observations.

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The main results are:

- 1) *Optimal local and pooled decisions.* We derive the optimal policies, posterior variances, and team losses, including the effect of the disagreement penalty (Proposition 1).
- 2) *An exact delay crossover.* We identify precisely when fresh local decisions have lower team loss than decisions based only on a complete delayed pool (Corollary 1).
- 3) *Three numbers suffice for hybrid decisions.* We derive the optimal combined policy and show that its pooling gain decays exactly exponentially with age. Its posterior variance obeys a matching interpolation law (Theorem 1).
- 4) *The value of local storage.* We quantify the loss from discarding the old local reading. Keeping it strictly improves performance at every positive finite pool age (Proposition 2).
- 5) *Unequal sensor quality.* A weighted pooled average gives the same exponential law when sensors have different error variances (Proposition 3).
- 6) *Different temporal rates.* For independent signal components, the total pooling gain is a sum of exponentials weighted by their decision value (Corollary 2).
- 7) *When and how often to communicate.* We derive the communication-price and latency thresholds for choosing no refreshes. Below the price threshold, the unique optimal period is optimal among all schedules independent of the observations (Theorem 2).

Each refresh supplies a finite total improvement while agents continue using fresh local observations. Once its price reaches that lifetime benefit, local operation is optimal.

Static team theory gives optimality conditions for agents with different observations [1], and common-information methods separate shared and private information [2]. Age of information [3], nonlinear age penalties [4], and remote estimation [5] connect freshness to performance. Ballotta and coauthors study communication latency in distributed control [6] and its effect on the spatial locality of optimal proportional feedback [7].

Kar and Varshney study sensor collaboration and sampling of an Ornstein–Uhlenbeck signal under a power budget [8]. Nayyar, Başar, Teneketzis, and Veeravalli jointly optimize communication and remote estimation under communication costs and energy constraints [9]. Their transmission decisions can depend on sensor observations. Our setting gives every agent fresh private observations between pooled updates and

penalizes disagreement among decisions. The exact age law quantifies the value of the shared update in this team, and the refresh theorem optimizes schedules chosen independently of observed values.

A longer version of this paper [10] develops a broader theory of information architecture, covering local versus delayed-global information, spatial predictability, and neighborhood-radius selection. This letter gives a self-contained analysis of common-signal Gaussian teams, including all proofs of the pooling and refresh results.

II. THE TEAM AND ITS INFORMATION

A. Signal, observations, and objective

There are $n \geq 2$ agents with scalar actions $u_i(t)$. The common signal $X(t)$ and sensor disturbances $E_i(t)$ are mutually independent, zero-mean, stationary Gaussian processes with

$$\begin{aligned} \text{Cov}(X(t), X(s)) &= ae^{-\lambda|t-s|}, \\ \text{Cov}(E_i(t), E_i(s)) &= be^{-\lambda|t-s|}, \end{aligned} \quad (1)$$

where $a, b, \lambda > 0$. Each is an Ornstein–Uhlenbeck process [11]. Agent i observes

$$Y_i(t) = X(t) + E_i(t), \quad \bar{Y}(t) = \frac{1}{n} \sum_i Y_i(t). \quad (2)$$

The disturbance represents temporally correlated local measurement error. The common rate in (1) makes the sensor vector evolve by one scalar prediction factor.

At each time the team minimizes the expected loss

$$\ell(u, X) = \frac{1}{n} \sum_i (u_i - X)^2 + \frac{\kappa}{n} \sum_i (u_i - \bar{u})^2, \quad \bar{u} = \frac{1}{n} \sum_i u_i, \quad (3)$$

where $\kappa \geq 0$. The terms price tracking error and disagreement. Actions affect neither the processes nor the observations. The problem is a sequence of static team decisions. All measurable policies with finite second moment are admissible when they satisfy the stated information restrictions.

B. Architectures and stored information

Write $\mathcal{F}_i(t) = \sigma\{Y_i(r) : r \leq t\}$ for the local history and $\mathcal{F}(t) = \bigvee_i \mathcal{F}_i(t)$ for the full sensor history. Fresh-local operation gives agent i the information $\mathcal{F}_i(t)$. Fresh-pooled operation gives every agent $\mathcal{F}(t)$. At pooling time $s = t - \tau$, delayed-pool-only operation gives every agent $\mathcal{F}(s)$ and uses no measurements taken after s . The hybrid architecture retains the local history through t and has information

$$\mathcal{H}_i(t; s) = \mathcal{F}_i(t) \vee \sigma\{\bar{Y}(r) : r \leq s\}. \quad (4)$$

Its pool is τ time units old and its local information is current. Let J_L , J_P , $J_D(\tau)$, and $J_H(\tau)$ be the respective minimum expected losses. Section III shows that the hybrid optimum needs only $Y_i(t)$, the retained sample $Y_i(s)$, and the latest pooled mean $\bar{Y}(s)$. The same value therefore applies when pooled information arrives only at selected refresh times. Table I summarizes the data used by the policies, including a restricted-memory comparison in Section III-C.

TABLE I

DATA SUFFICIENT TO IMPLEMENT THE OPTIMAL POLICIES. THE LAST ROW RESTRICTS EACH DEVICE TO ITS CURRENT LOCAL READING.

Architecture	Local data used	Shared scalar
Fresh-local	$Y_i(t)$	None
Fresh-pooled	None	$\bar{Y}(t)$
Delayed pool only	None	$\bar{Y}(s)$
Hybrid	$Y_i(t), Y_i(s)$	$\bar{Y}(s)$
Current reading + pool	$Y_i(t)$	$\bar{Y}(s)$

We call $J_L - J_H(\tau)$ the *coordination value* of the delayed pool. It includes reductions in tracking error and disagreement. Its age-zero value is $\Delta = J_L - J_P$.

C. Optimality and the fresh endpoints

The next condition establishes optimality without restricting policies to linear rules.

Lemma 1 (Team optimality). *For information fields $\mathcal{G}_i$, a feasible policy u is optimal if and only if*

$$(1 + \kappa)u_i - \kappa\mathbb{E}[\bar{u} | \mathcal{G}_i] = \mathbb{E}[X | \mathcal{G}_i], \quad i = 1, \dots, n. \quad (5)$$

The optimal policy is unique up to almost-sure equality.

Proof. For a feasible perturbation w , expansion of (3) gives the linear term $2n^{-1} \sum_i \mathbb{E}[w_i((1 + \kappa)u_i - \kappa\bar{u} - X)]$. It vanishes for every w exactly when (5) holds. The remaining term is $n^{-1} \sum_i \mathbb{E}[w_i^2] + \kappa n^{-1} \sum_i \mathbb{E}[(w_i - \bar{w})^2]$, which is positive for every nonzero perturbation. This proves necessity, sufficiency, and uniqueness. $\square$

Define

$$v = a + \frac{b}{n}, \quad d = a + b \left[1 + \kappa \left(1 - \frac{1}{n} \right) \right]. \quad (6)$$

Proposition 1 (Fresh endpoints). *The posterior variances are*

$$\begin{aligned} P_1 &= \text{Var}(X(t) | \mathcal{F}_1(t)) = \frac{ab}{a+b}, \\ P_n &= \text{Var}(X(t) | \mathcal{F}(t)) = \frac{ab}{b+na}. \end{aligned} \quad (7)$$

The optimal policies and costs are

$$u_i^L(t) = \frac{a}{d} Y_i(t), \quad J_L = a - \frac{a^2}{d}, \quad (8)$$

$$u_i^P(t) = \frac{a}{v} \bar{Y}(t), \quad J_P = P_n = a - \frac{a^2}{v}. \quad (9)$$

In particular,

$$\Delta = \frac{a^2(d-v)}{vd} > 0. \quad (10)$$

Proof. For any $r \leq t$, the residual $X(t) - aY_i(t)/(a+b)$ has zero covariance with $Y_i(r)$. Gaussianity makes it independent of the local history. Similarly,

$$\eta(t) = X(t) - \frac{a}{v} \bar{Y}(t) \quad (11)$$

has zero covariance with $Y_j(r)$ for every j and every r . It is independent of the full sensor process. Computing the residual variances gives (7) and the corresponding conditional means.

For $u_i = kY_i(t)$, $\mathbb{E}[\bar{u} \mid \mathcal{F}_i(t)] = kvY_i(t)/(a+b)$. Equation (5) then gives $k[(1+\kappa)(a+b) - \kappa v] = a$. The bracket equals d , proving the local policy. Its expected loss is $a - 2ak + dk^2 = a - a^2/d$. With pooled information, every agent uses the same conditional mean of $X(t)$, with zero disagreement and loss P_n . Lemma 1 proves optimality. Finally, $d - v = b(1+\kappa)(1-1/n) > 0$ gives (10). $\square$

When $\kappa > 0$, the local action has smaller magnitude than the local posterior mean. The team trades some individual tracking accuracy for less disagreement. Specifically,

$$J_L - P_1 = \frac{\kappa a^2 b(1-1/n)}{(a+b)d}. \quad (12)$$

Pooling removes this additional cost and reduces posterior variance.

III. AN EXACT LAW FOR DELAYED POOLING

A. When local decisions outperform pooling

Pooling provides a common estimate from all sensors. Its age determines whether that estimate supports better decisions than current local measurements.

Corollary 1 (The delay crossover). *Under delayed-pool-only information, the optimal policy and loss are*

$$u_i^D(t) = e^{-\lambda\tau} \frac{a}{v} \bar{Y}(s), \quad J_D(\tau) = a - \frac{a^2}{v} e^{-2\lambda\tau}. \quad (13)$$

Fresh-local operation has strictly smaller team loss precisely when

$$\tau > \tau_{\text{cross}} := \frac{1}{2\lambda} \log \frac{d}{v}. \quad (14)$$

The two costs agree at $\tau = \tau_{\text{cross}}$.

Proof. The Markov property and Proposition 1 give $\mathbb{E}[X(t) \mid \mathcal{F}(s)] = e^{-\lambda\tau} (a/v) \bar{Y}(s)$. Taking this common conditional mean minimizes tracking loss and gives zero disagreement, hence also minimizes the team loss. Its error variance is (13). Comparing with $J_L = a - a^2/d$ gives $J_L < J_D$ exactly when $e^{-2\lambda\tau} < v/d$, which is (14). $\square$

This crossover compares decision losses alone. Stronger disagreement penalties increase d and therefore increase the delay that pooling can tolerate. Faster temporal change increases λ and shortens that tolerance.

B. Retaining fresh local information

The hybrid keeps current local measurements available while the pool ages. The next theorem gives its exact improvement over local operation.

Theorem 1 (Hybrid policy and exponential value). *Let $s = t - \tau$ and $\rho = e^{-\lambda\tau}$. Under (1)–(4), the optimal hybrid policy is*

$$u_i^H(t) = \rho \frac{a}{v} \bar{Y}(s) + \frac{a}{d} (Y_i(t) - \rho Y_i(s)). \quad (15)$$

It attains the optimum even if every agent receives all sensor histories through s . The posterior variance $P_H(\tau) = \text{Var}(X(t) \mid \mathcal{H}_i(t; s))$ and team cost satisfy

$$P_H(\tau) = \rho^2 P_n + (1 - \rho^2) P_1, \quad (16)$$

$$J_H(\tau) = \rho^2 J_P + (1 - \rho^2) J_L. \quad (17)$$

Consequently,

$$J_L - J_H(\tau) = \Delta e^{-2\lambda\tau}. \quad (18)$$

Proof. For $\tau = 0$, (15) is the pooled policy. Suppose $\tau > 0$ and enlarge each agent's information to $\mathcal{F}(s) \vee \mathcal{F}_i(t)$. Set

$$Z_i = Y_i(t) - \rho Y_i(s), \quad h = \frac{a}{v}, \quad k = \frac{a}{d}. \quad (19)$$

The Gaussian Markov property makes $Z = (Z_1, \dots, Z_n)$ independent of $\mathcal{F}(s)$, with covariance $(1 - \rho^2)(a\mathbf{1}\mathbf{1}^\top + bI)$. Equation (11) gives

$$X(t) = \rho h \bar{Y}(s) + h \bar{Z} + \eta(t). \quad (20)$$

The last term is independent of all sensor data and has variance P_n .

We first justify using only the endpoint of each private innovation path. For $s < r < t$, let $Z_i(r) = Y_i(r) - e^{-\lambda(r-s)} Y_i(s)$ and

$$c(r) = \frac{e^{-\lambda(t-r)} (1 - e^{-2\lambda(r-s)})}{1 - e^{-2\lambda(t-s)}}. \quad (21)$$

A covariance calculation shows that every $Z_i(r) - c(r)Z_i$ is uncorrelated with the entire vector Z . These bridge variables are also independent of the old sensor history and of $\eta(t)$. Joint Gaussianity makes the bridges jointly independent of the variables that determine the current loss. Averaging the policies over the bridges preserves feasibility and cannot increase a convex loss. Hence private paths add no value beyond Z_i .

Condition on $\mathcal{F}(s)$. The first term in (20) is common knowledge. After subtracting it from every action, the residual team observes Z_i and tracks $h\bar{Z} + \eta(t)$. Gaussian regression gives

$$\begin{aligned} \mathbb{E}[h\bar{Z} + \eta(t) \mid Z_i] &= \frac{a}{a+b} Z_i, \\ \mathbb{E}[k\bar{Z} \mid Z_i] &= \frac{kv}{a+b} Z_i. \end{aligned}$$

Since $(1+\kappa)(a+b) - \kappa v = d$, Lemma 1 proves that kZ_i is the optimal correction. The resulting policy (15) is feasible under (4), so it also attains the lower bound from complete delayed sharing.

At one time, $X = h\bar{Y} + \eta$ separates the local cost into the irreducible term P_n and the remaining cost $J_L - P_n$. Replacing Y by Z scales the latter quadratic cost by $1 - \rho^2$. Thus $J_H = P_n + (1 - \rho^2)(J_L - P_n)$, giving (17). The posterior mean under the enlarged information is $\rho h \bar{Y}(s) + aZ_i/(a+b)$. It is feasible under (4) and has error variance $P_n + (1 - \rho^2)(P_1 - P_n)$. This proves (16). Equation (18) follows by subtraction. $\square$

The first term in (15) predicts the signal from the pooled average. The second corrects it using the difference between the current local reading and its prediction from the stored reading. Evaluation needs the pooled average, the local reading stored at that pooling time, and the current reading. Earlier data add no decision value.

Retaining at least a fraction $\alpha \in (0, 1)$ of the fresh-pooling value requires

$$\tau \leq \frac{\log(1/\alpha)}{2\lambda}. \quad (22)$$

This age budget is independent of n , a , b , and κ . Those parameters determine the amount Δ of value available to preserve. The value half-life is $\log 2/(2\lambda)$.

C. The value of retaining a local sample

The retained sample permits an exact local innovation calculation. To quantify its value, consider a device that uses only its current reading and the last pooled mean. Its information is $\sigma\{Y_i(t), \bar{Y}(s)\}$, and let its optimal team cost be $J_C(\tau)$.

Proposition 2 (Current reading and old pool). *Put $z = e^{-2\lambda\tau}$. The coordination value of this architecture is*

$$J_L - J_C(\tau) = \Delta \frac{z(d-v)}{d-vz}. \quad (23)$$

For every $0 < \tau < \infty$, it is strictly smaller than the retained-sample value Δz .

Proof. Let $M = \bar{Y}(s)$, $R_i = Y_i(t) - \sqrt{z}M$, and $\tilde{X} = X(t) - \sqrt{z}(a/v)M$. Gaussian regression makes $(\tilde{X}, R)$ independent of M , with

$$\begin{aligned} \text{Cov}(\tilde{X}, R_i) &= a(1-z), \\ \text{Var}(R_i) &= a + b - zv, \quad \text{Cov}(R_i, \bar{R}) = v(1-z). \end{aligned}$$

Conditioning on M and applying Lemma 1 gives the unique optimal policy

$$u_i = \sqrt{z} \frac{a}{v} M + \frac{a(1-z)}{d-zv} R_i. \quad (24)$$

Here $d - zv > 0$. Substitution into the conditional quadratic cost yields

$$J_C = a - \frac{a^2 z}{v} - \frac{a^2(1-z)^2}{d-zv}. \quad (25)$$

Subtracting this expression from J_L and using (10) gives (23). For $0 < z < 1$, $(d-v)/(d-zv) < 1$, which proves the strict comparison. $\square$

Keeping one local sample therefore improves the value of the same broadcast without additional communication.

IV. UNEQUAL SENSOR QUALITY AND TEMPORAL RATES

A. Sensors with different disturbance variances

The scalar pooling law also permits unequal sensor quality. Let $\text{Var}(E_i(t)) = b_i > 0$, with the common rate λ unchanged, and define

$$C_i = (1+\kappa)a + b_i[1+\kappa(1-1/n)], \quad h_0 = \frac{1}{n} \sum_i C_i^{-1}. \quad (26)$$

Proposition 3 (Unequal sensor quality). *The fresh-local coefficients, fresh-pooled posterior, and costs are*

$$\begin{aligned} k_i &= \frac{a}{C_i(1-\kappa a h_0)}, & J_L &= a - \frac{a^2 h_0}{1-\kappa a h_0}, \\ P &= \left(a^{-1} + \sum_i b_i^{-1} \right)^{-1}, & \mu_P(t) &= P \sum_i \frac{Y_i(t)}{b_i}, \end{aligned} \quad (27)$$

with $u_i^L = k_i Y_i$ and $J_P = P$. Broadcasting the single scalar $\mu_P(s)$ gives the hybrid optimum

$$u_i^H(t) = \rho \mu_P(s) + k_i[Y_i(t) - \rho Y_i(s)], \quad (28)$$

and $J_H = \rho^2 J_P + (1 - \rho^2) J_L$.

Proof. Substitution of $u_i = k_i Y_i$ into (5) gives $C_i k_i - \kappa a \bar{k} = a$, where $\bar{k} = n^{-1} \sum_i k_i$. Averaging $k_i = (a + \kappa a \bar{k})/C_i$ solves for $\bar{k}$ and gives (27). The denominator is positive because $C_i > (1 + \kappa)a$. At the optimum the quadratic term equals $a\bar{k}$, so $J_L = a - a\bar{k}$. Gaussian conditioning gives the displayed P and μ_P .

The residual $X(t) - \mu_P(t)$ is independent of the entire sensor process, since all temporal covariances share the same factor. Innovations over $[s, t]$ have covariance $(1 - \rho^2)(a\mathbf{1}\mathbf{1}^\top + \text{diag}(b_i))$. The bridge argument and conditional team decomposition in Theorem 1 therefore apply with these covariances and the coefficients k_i . They give (28) and the stated cost law. $\square$

B. Independent components with different rates

A task may contain components that change on different time scales. Suppose each agent selects $(u_{i1}, \dots, u_{iM})$ and the loss is $\sum_m w_m \ell_m$, with $w_m > 0$. Component m follows the model above with parameters $(a_m, b_m, \kappa_m, \lambda_m)$, and the component processes are mutually independent. Within each component, the common signal and its sensor disturbances share the rate λ_m . The rates may differ across components. Define $J_{L,m}$, $J_{P,m}$, and Δ_m by Proposition 1.

Corollary 2 (Additive coordination value). *If summary m has age τ_m , its optimal policy is (15) with the corresponding parameters. The total coordination value is*

$$J_L^{\text{tot}} - J_H^{\text{tot}} = \sum_m w_m \Delta_m e^{-2\lambda_m \tau_m}, \quad J_L^{\text{tot}} = \sum_m w_m J_{L,m}. \quad (29)$$

Proof. For component m , observations of the other components are independent of its signal and observations. Averaging a policy over those observations preserves the componentwise information constraints. Convexity ensures that this cannot increase its loss. Optimization therefore separates across components. Apply Theorem 1 to each and sum with weights w_m . $\square$

For a common age τ , write the gain in (29) as $V(\tau)$. Its logarithmic decay rate is

$$-\frac{V'(\tau)}{V(\tau)} = \frac{\sum_m 2\lambda_m w_m \Delta_m e^{-2\lambda_m \tau}}{\sum_m w_m \Delta_m e^{-2\lambda_m \tau}}. \quad (30)$$

This is a weighted average of the component rates $2\lambda_m$. Slowly changing components account for a larger share of the remaining value as the pool ages.

V. OPTIMAL REFRESHING UNDER A COMMUNICATION COST

A communication service delivers each pooled vector after a fixed latency $\delta \geq 0$. It supports the chosen transmission times without queueing. On reception, the pool's age resets to δ . Agents retain local measurements from the associated sampling time. During transmission these samples can be kept in a timestamped buffer. Each vector costs $c > 0$ in units of accumulated team loss. The refresh schedule is independent of all signal and disturbance processes. This includes deterministic schedules and independent randomized schedules.

For joint refreshing of all M components, set

$$\begin{aligned} \gamma_m &= 2\lambda_m, & A_m &= w_m \Delta_m e^{-\gamma_m \delta}, \\ B(T) &= \sum_m \frac{A_m}{\gamma_m} (1 - e^{-\gamma_m T}). \end{aligned} \quad (31)$$

Here $B(T)$ is the gain accumulated in duration T after a reception. Periodic refreshing with period $T > 0$ has average cost

$$C(T) = J_L^{\text{tot}} + \frac{c - B(T)}{T}. \quad (32)$$

For a general schedule, the objective is the limiting upper average of expected integrated team loss plus c times the number of transmissions. With no refreshes it equals J_L^{tot} .

Theorem 2 (Optimal refresh period). *Define*

$$c_{\text{crit}} = \sum_m \frac{A_m}{\gamma_m}. \quad (33)$$

If $c \geq c_{\text{crit}}$, the schedule with no refreshes attains the minimum average cost J_L^{tot} . If $0 < c < c_{\text{crit}}$, periodic refreshing is optimal among all locally finite schedules independent of the sensor processes. Its period is the unique positive solution of

$$\sum_m \frac{A_m}{\gamma_m} [1 - (1 + \gamma_m T^*) e^{-\gamma_m T^*}] = c. \quad (34)$$

The minimum average cost is

$$C(T^*) = J_L^{\text{tot}} - \sum_m A_m e^{-\gamma_m T^*}. \quad (35)$$

Proof. Put $V_\delta(T) = B'(T) = \sum_m A_m e^{-\gamma_m T}$. Differentiating (32) gives

$$C'(T) = \frac{H(T) - c}{T^2}, \quad H(T) = B(T) - TV_\delta(T). \quad (36)$$

Since $H'(T) = -TV'_\delta(T) > 0$, H increases strictly from zero to c_{crit} . For $0 < c < c_{\text{crit}}$, C decreases up to the unique root $H(T^*) = c$ and then increases. Also, $C(T) \rightarrow \infty$ as $T \downarrow 0$ and $C(T) \rightarrow J_L^{\text{tot}}$ as $T \rightarrow \infty$. These facts prove the unique minimizing period and (34). At the root, $c - B(T^*) = -T^* V_\delta(T^*)$, proving (35). If $c \geq c_{\text{crit}}$, then $c - B(T) \geq 0$ for every finite T .

To compare general schedules, set $g = \min\{0, \inf_{T>0} (c - B(T))/T\}$. Condition on any realization of an independent reception schedule. Each complete reception interval T_j contributes at least gT_j relative to fresh-local operation after assigning one communication cost to it. The last incomplete interval contributes at least $-c_{\text{crit}}$, which bounds the gain over its entire lifetime. The initial interval before any reception has zero gain. Because $g \leq 0$, the total cost through time R is at least

$$R(J_L^{\text{tot}} + g) - c_{\text{crit}}.$$

Charging transmissions at generation preserves this lower bound because the number sent is at least the number received. Divide by R and let $R \rightarrow \infty$. The bound holds for every schedule realization and hence for independent randomized schedules. Periodic operation at T^* , or no communication in the other case, attains it. $\square$

For one component, $x = 2\lambda T^*$ and $\chi = c/c_{\text{crit}}$ reduce the period equation to

$$1 - (1 + x)e^{-x} = \chi, \quad 0 < \chi < 1. \quad (37)$$

The left side increases strictly, so bisection determines the period. As the price approaches its threshold, the interval diverges. For a single component of unit weight, the no-refresh condition has the equivalent latency form

$$\delta \geq \delta_{\text{stop}} := \max \left\{ 0, \frac{1}{2\lambda} \log \frac{\Delta}{2\lambda c} \right\}. \quad (38)$$

This follows by solving $c \geq \Delta e^{-2\lambda\delta}/(2\lambda)$ for δ . At or above this threshold, every finite periodic refresh schedule has larger total cost than local operation. No refreshes also attain the optimum over the full independent-schedule class in Theorem 2. If the service requires intervals of at least $T_{\min} > 0$, the derivative sign in (36) gives the constrained period $\max\{T_{\min}, T^*\}$ below the same threshold.

VI. NUMERICAL ILLUSTRATION

Consider $n = 8$, $a = b = 1$, $\kappa = 1$, and $\lambda = 1$. Then $P_1 = 0.5$, $J_P = P_n = 1/9$, $J_L = 15/23$, and $\Delta = 112/207 \simeq 0.5411$. The delay crossover is $\lambda\tau_{\text{cross}} \simeq 0.4691$. At one coherence time, $\lambda\tau = 1$, delayed-pool-only loss is 0.8797, compared with 0.6522 locally. Local operation therefore reduces team loss by 25.9%, with no communication price included. At age $\tau = 0.5$, the hybrid retains $e^{-1} \simeq 36.8\%$ of the fresh-pooling value and has loss 0.4531. Figure 1 uses 120,000 independent stationary Gaussian sample pairs per marker. We also solve the Gaussian team normal equations directly from observation covariance matrices and compare the resulting policies and costs with the formulas. Across 126 systems, the maximum cost discrepancy is 5.3×10^{-15} .

For latency $\delta = 0.1$, the critical communication cost is $\Delta e^{-0.2}/2 \simeq 0.2215$. Figure 2 shows the effect of communication price. At half this cost, (37) gives $2\lambda T^* \simeq 1.6783$, hence $T^* \simeq 0.8392$. The average cost is approximately 0.5695.

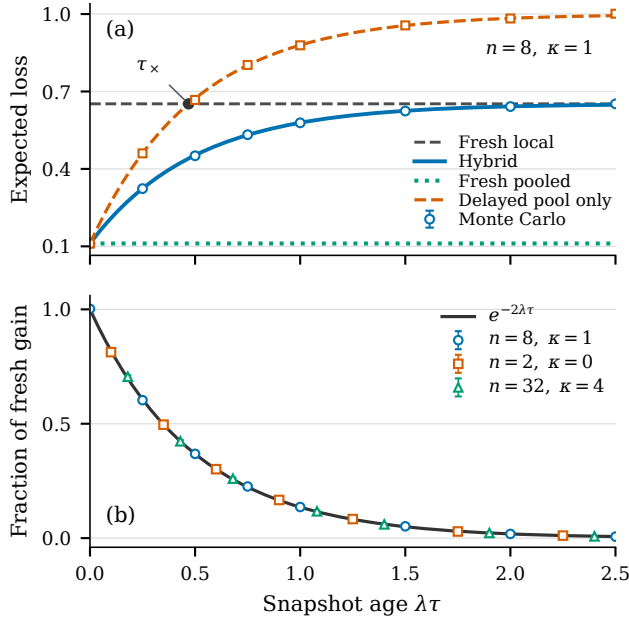


Fig. 1. The delay crossover and hybrid value. A delayed pool used alone becomes worse than fresh-local operation beyond τ_{cross} . The hybrid retains fresh local information and improves on local operation. Its normalized gain follows the exponential law. Markers use stationary Gaussian sample pairs.

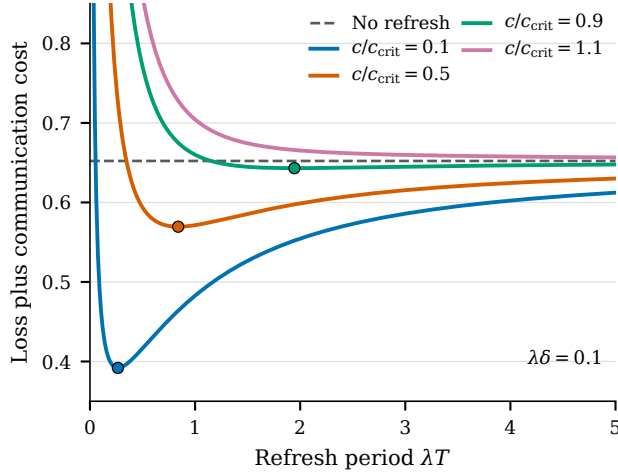


Fig. 2. Average team loss plus communication cost with periodic refreshing. Below the critical communication cost, each curve has a unique finite minimizer. Above it, fresh-local operation attains the minimum average cost.

To illustrate unequal rates, add an independent component with $a_2 = 0.4$, $b_2 = 0.8$, and $\lambda_2 = 0.12$, keeping $n = 8$, $\kappa_2 = 1$, and unit weights for both components. At latency 0.1, the slow component supplies 34.2% of the value at reception and 81.2% of the integrated gain c_{crit} . The joint threshold is 1.1807. At half this threshold, the optimal period is 5.5513 and the total average cost is 0.9072. Thus a component's contribution to the refresh decision depends on both the magnitude and the duration of its useful information.

VII. CONCLUSION

Fresh local readings yield lower team loss than a complete delayed pool once its age exceeds an explicit threshold. An agent that also uses fresh local data can match the performance of complete delayed sharing using only three numbers: its current reading, its reading at the pooling time, and the pooled average. The improvement over local operation decays exponentially. Its finite lifetime value determines when communication is worth its price and how often to refresh. The pooling law also extends to unequal sensor quality and independent components with different temporal rates. The refresh result assumes fixed latency and schedules independent of the observations. Different signal and sensor-error rates within a component, or observation-dependent scheduling, require further analysis.

Simulation code, extended proofs, and proof verification reports will be posted at <https://github.com/anrgusc/TINA>.

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